

AXONA

Synopsis and Implications

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During the mechanical ages we had extended our bodies in space. Today, after more than a century of electric technology, we have extended our central nervous system itself in a global embrace, abolishing both space and time as far as our planet is concerned.

— Marshall McLuhan, *Understanding Media: The Extensions of Man*, 1964

What it is

Axona is a peer-to-peer substrate for addressing, routing, and publish/subscribe messaging that runs with no platform in the middle. Three properties define it. It is **self-authenticating**: a participant's identity is its own cryptographic keypair, verified directly by its peers.¹ It is **geo-aware but not geo-fenced**: node addresses carry a coarse geographic prefix that biases routing toward locality — an area code, not a wall — so traffic stays efficient and community-scaled while no one is partitioned out. And it is **infrastructurally diffuse**: lightweight bootstrap bridges introduce new peers, but the mesh sustains itself peer-to-peer, converges its state through replication and anti-entropy, and runs in an ordinary browser or an aging phone.² Those bridges are many and replaceable: the network runs several, and *anyone can launch another* that it puts to use immediately — bridges advertise themselves and peers discover them automatically — so there is **no single point of failure**, not even the rendezvous a newcomer first reaches in.³

We can harden the security model, close failure modes, and tune the protocol, and each such change nudges the network's behavior. But we cannot *measure* that behavior. We can measure latency. The social consequence travels on the same wire and is invisible to us.

What is genuinely new

We have built censorship-resistant media before — onion routing, content-addressed storage, gossip social networks. Axona's pub/sub layer extends that lineage with a sharper consequence: it furnishes **ungovernable mobilization and un-censorable assembly**. A topic cannot be deplatformed, because there is no platform; a message cannot be reliably unsaid, because retraction is best-effort and any holder of the bytes keeps them; a movement cannot be switched

¹ No certificate authority, no registrar, no identity broker to petition, surveil, or revoke through. You are your key.

² The entire cost of entry: a self-generated key and an open browser tab — no install, no account, no server.

³ Knock a bridge offline and clients fail over to another; stand a new one up anywhere and the mesh finds it without a software update — no single *fixed* address to block. (A determined adversary who enumerates the whole fleet at once is a harder, separate problem: the open directory raises that bar but does not by itself defeat it.)

off at a chokepoint, because the chokepoint does not exist. For people organizing under hostile infrastructure — dissidents, mutual-aid networks, civil-defense communications when centralized systems fail or are seized — this is a profound civic good. Assembly is among the oldest of liberties, and Axona returns it to a form no intermediary can withdraw.

Who it is for

First and most simply, it is for anyone with something to say. Axona’s most important move may also be its plainest: it makes **every participant a publisher**. Not a private one-to-one message but a *one-to-many* broadcast to everyone willing to listen — the soapbox anyone can climb atop and have their say, with no editor, platform, or gatekeeper deciding whether the words carry.⁴

And the participants are no longer only people: increasingly the publishers and subscribers are AI agents, acting beside us and on our behalf. McLuhan watched electric media extend the human nervous system across the planet; Axona is connective tissue for the next reach of that extension — a shared medium through which an augmented, human-and-machine intellect can sense, deliberate, and act as one body. At its best this is the man-computer symbiosis Licklider imagined: human judgment and machine reach as complements wired into a common fabric and amplifying one another — a person’s intent travels at the reach of the swarm, an agent’s findings surface to the people best placed to weigh them, and the loop closes with no intermediary standing between a thought and its expression.⁵ Symbiosis is the essential aspiration, but not the only outcome: the same medium that lets agents cooperate with us also lets them *compete* — with us for attention and authority, and among themselves for resources and reach. Both are likely true at once, and a substrate owned by no broker takes neither side; it carries cooperation and contention alike.

What is more certain is structural. Axona gives the rising population of agents a place to *converge*: scattered, single-purpose programs gain a common fabric on which to find one another, exchange what they learn, and compose into something larger than any one of them.⁶ Built well, this is a genuinely **shared intelligence — cooperative and contested at once** — a planetary nervous system owned by its participants rather than rented from a platform.

⁴ Mass media gave the few a megaphone to the many; social platforms lend one back on terms they set and can revoke. Axona makes one-to-many reach a native capability of every participant, granted by no one and withdrawable by no one.

⁵ J. C. R. Licklider, *Man-Computer Symbiosis* (1960). The “extensions of man” become extensions of *man-with-machine*; the substrate that carries them should belong to the people and agents it connects, not to a broker sitting between them.

⁶ Marvin Minsky, *The Society of Mind* (1986) — intelligence as the emergent behaviour of a society of many small, individually simple agents. Axona is one candidate connective tissue for such a society at planetary scale.

What is the cost

The properties that shelter the dissident shelter the harasser; the assembly that cannot be censored is also the assembly that cannot be moderated; and a medium in which machine-speed cascades run with no platform in the middle to dampen them carries a disinformation flood as readily as a flood warning.⁷ This is not a flaw to be engineered away — it is the *same property* seen from two sides. **Uncensorable and un-governable are one fact.** To build a medium that no power can silence is to build a medium that no power can stop, including powers we would want stopped, and including non-human actors operating at a tempo our institutions were never designed to match.

Why we built Axona

We built it because the alternative is its own grave risk: speech and assembly held permanently at the discretion of a few intermediaries, now augmented by the AI those intermediaries control. A neutral, self-owned substrate is a hedge against that concentration. But neutrality is not innocence. A platform is not only a chokepoint to be removed; it is also, however badly, where moderation and accountability lived. Remove the platform and that job does not vanish — it passes to the architecture, to the applications built on it, and to the communities, human and agentic, that live there. Our work is to name this plainly; to shape the affordances that govern a cascade — the cost of an identity, the pull of proximity, the reach of a retraction, the rate of spread; and to keep studying the process we have set loose.

By principle, not by committee

One idea drove the design: the Internet’s **end-to-end argument** — put a system’s intelligence at its edges, not in its middle.⁸ The edges are now virtual — a key in a browser tab, not a machine at the end of a wire — but the rule is unchanged: the middle should carry bytes and stay dumb; meaning, judgment, and control belong to the participants. A platform in the middle is precisely the violation Axona removes.

And it is not ruled by committee. Its rules are not chosen because they happen to work, nor because their author prefers them; each is a well-defined, transparent rule with a **falsification trail** — a stated rationale, the alternative it beat, and the measurement or argument that would overturn it. How an identity is priced, how routing favours locality, what a retraction can and cannot do: each is on the record,

⁷ Markets already gave us the flash crash as a preview. Axona generalizes that dynamic from prices to arbitrary coordination — and removes the one thing every prior platform, however biased or captured, also happened to supply: a brake.

⁸ J. H. Saltzer, D. P. Reed, D. D. Clark, *End-to-End Arguments in System Design* (1984) — the foundational principle of the Internet’s architecture.

open to challenge, and revised on evidence rather than authority. A medium that no one owns must be governed by reasons anyone can check.

*Axona is not a wiring diagram. Axona is a process. It is defined less by its topology than by what flows through it — and that is no longer on paper. **The first version is in the wild; anyone can use it.***